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Palos Verdes Reef: a test of artificial reef design criteria to increase regional fish production in southern California

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ABSTRACT.—The first offshore and in situ restoration rocky reef in California, the Palos Verdes Reef (PVR), was designed to maximize the secondary production of fishes. The overall project goal was to restore 35.2 ha of sediment impacted or buried rocky reef utilizing 52,729 mt of granitic quarry rock modules distributed strategically with a footprint of 3.2 ha. Module design, modeled after a nearby highly productive natural reef feature, incorporated relief and rock size criteria to theoretically withstand sedimentation and turbidity impacts. In a first for temperate reef ecosystems, reefing modules were placed adjacent to the natural reef. Further design criteria included maximization of ecotonal perimeter and soft-bottom halos increasing niche and resource availability. Postconstruction, we documented the relocation of resident reef fishes from surrounding reefs to the newly created habitat. From 2021 to 2023 for reef associated fishes, standing biomass and secondary annual production increased across all modules: total fish biomass increased from 223 to 3286 kg (1372%) and production increased 59 to 690 kg yr^{-1} (1062%). Halo biomass density increased; from 6.9 to 22.5 g m^{-2} and production density from 1.8 to 6.0 g m^{-2} yr^{-1}. Due to their relatively large area, overall halo production was nearly equivalent to module production. Over the entire restoration site, fish biomass increased 96% and production increased 93%, demonstrating its carrying and productive capacity was almost doubled during the initial three-year period postconstruction PVR modules. These metrics of strong success occurred while the system is still developing, despite declining biomass and production at reference sites.

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Well-designed artificial reefs (ARs), typically designed with replicated reefing modules, improve localized and regional production (Pondella et al. 2022). Lessons learned from existing structures demonstrate that reefing projects, which add hard

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substrate to soft bottom habitats, produce localized increases in fish and invertebrate density and biomass (Johnson et al. 1994, Santos and Monteiro 1998, Steimle et al. 2002, Birt et al. 2024). This is particularly effective when these structures support multiple ontogenetic life stages for fishes to maintain high levels of secondary somatic production across their size range and gonadal production (i.e., reproductive output) of larger adults (Claisse et al. 2014, 2019, Pondella et al. 2015). Perhaps the best example of a regional-scale increase is from Portugal, where ARs increased regional carrying capacity *K* based upon fishery landings (Roa-Ureta et al. 2019). It is not surprising, due to habitat challenges (i.e., associated losses, deterioration and overfishing), ARs are increasingly being designed for fisheries enhancement (Lemoine et al. 2019). While there is ample evidence that properly designed ARs increase localized fish production, and in some cases regional fish production, how to optimize this production maximizing the use of limited resources is a critical endeavor.

Traditionally, ARs placed on soft bottom habitats create an island effect, which increases localized production by providing habitat that attracts and recruits new fish assemblages (Polovina and Sakai 1989, Johnson et al. 1994, Steimle et al. 2002). Reefs add three-dimensional structural habitat increasing fish density, diversity and production over the soft bottom habitats (Bond et al. 1999, Pondella et al. 2006, 2022). Beyond the simplistic addition of hard substrate, reefs have an ecotonal perimeter (the transition between hard and soft bottom habitats), which provides additional niche availability, as some taxa primarily occupy this transitional zone, and an interplay among hard and soft bottom-associated communities (Anderson et al. 1989). The effect these structures have beyond their ecotonal perimeters to the surrounding soft bottom habitats is known as the "halo" effect (Reeds et al. 2018, Rosemond et al. 2018). Demersal fishes use the reef for shelter and forage on the surrounding soft bottom infauna communities, which consume detritus from the reefs (Danovaro et al. 2002, Fabi et al. 2002). Depending upon the vagility of the reef fishes, these halos facilitate reef connectivity and allow soft bottom fishes to move through them, increasing species richness and diversity of the overall reef communities (Ebeling et al. 1980, Pondella and Allen 2000). Increased richness and niche availability are correlated with higher production, which is negatively affected by anthropogenic pressure (Mittelbach et al. 2001, Carmezim et al. 2022). Beyond the two-dimensional halos, reefs also create a three-dimensional dome effect, extending the halo into the water column, generating upwelling and facilitating feeding by planktivores (Zhang et al. 2021, Pondella et al. 2022). These reef-based planktivores shelter within these structures, feed in the water column, and provide a trophic transfer from planktonic flux to the reef community (Bray et al. 1981, Cresson et al. 2014a,b, Truong et al. 2017, Holland et al. 2021). Additionally, these reefs attract pelagic planktivores increasing localized reef production (Smith et al. 2016). There is a dynamic interplay among these reefing structures, soft bottom, and pelagic communities.

Optimal habitat characteristics (e.g., shelter, heterogeneity, relief, complexity, spacing), and orientation relative to current flow theoretically increase the scale of overall total secondary production (Pondella et al. 2022). Habitat loss and degradation, typically from anthropogenic stressors, limit fish richness and associated production (Carmezim et al. 2022). Theoretically, the addition of habitat designed with reef characteristics that support multiple ontogenetic stages (e.g., nursery habitats, shelter and spawning sites) and high species richness has the potential to increase

local and regional fish production beyond the physical footprint of the reef. Quality reef habitats increase recruitment, postsettlement survival, and are preferred by large fecund adults; their presence positively affects the surrounding ecosystem by the increased reproductive output, and this spillover process is a crucial feature of marine protected areas (Russ et al. 2004, Schmiing et al. 2017). The challenge is the restoration and corresponding increase in secondary fish production in a stressed e.g., pollution, sedimentation, overfished, etc.) ecosystem increasing regional carrying capacity beyond the localized reef footprint.

We tested these challenges utilizing a novel in situ restoration reefing project in southern California. The Palos Verdes Reef (PVR), was specifically designed utilizing reefing parameters that theoretically maximize secondary production of fishes such as relief, spacing, shelter, heterogeneity and complexity (Pondella et al. 2018, 2022, Williams et al. 2022). The setting has a long history of landslides and associated sedimentation (Kayen et al. 2002). Subtidal sedimentation causes reduced light penetration resulting in decreased primary production, and also smothers, scours, and/or buries rocky and coral reefs. PVR was designed to withstand these impacts plaguing temperate and tropical ecosystems (Airoldi 2003, Tuttle and Donahue 2022). In this project, the new reefs were built with discrete granitic quarry rock reefing modules, made up of heterogeneous submodules designed specifically to elevate the reef above the sedimentation, turbidity and scour processes. Utilizing spacing criteria based upon the known daily movement of southern California reef fishes (Johnson et al. 1994, Pondella et al. 2006), they were placed proximate to an existing natural reef, enabling the opportunity to assess increases in secondary fish production in the reef's footprint, its halo, and the connectivity to the larger reef system. We tested the utilization of artificial reef theory based upon the secondary production of reef fishes to restore lost and degraded habitat, a first for California and temperate rocky reef ecosystems.

Methods

Setting.—PVR was built within a giant kelp rocky reef ecosystem on Palos Verdes Peninsula in the County of Los Angeles, along one of the most populated coastlines in the world (Fig. 1). As such, this ecosystem is economically and ecologically valuable yet subjected to multiple typical anthropogenic stressors such as pollution and overfishing (Stull et al. 1987, Foster and Schiel 2010). It is also the location of multiple anthropogenically induced landslides. Immediately adjacent is Portuguese Bend, which has released 5.7–9.4 million metric tons from 1956–1999 prior to stabilization and Bunker Point, at the study site, where 6.5 ha slid in 1999 (Fig. 1) (Kayen et al. 2002, Pondella et al. 2018). As a result, much of the historic kelp-supported reef is buried or impacted by chronic sedimentation due to reef burial, scour, and associated turbidity (Pondella et al. 2018). Giant kelp immediately recruited to the shallow portions of modules (Fig. 2), and this reefing area is open to commercial and recreational fishing.

Reefing Design.—PVR consists of eighteen approximately 48 m × 16 m granitic quarry rock modules placed at 16.7–21.1 m seafloor depth. Within each module, submodules (i.e., piles) with varied vertical relief of 1–4 m were staggered, with high-relief submodules placed at the upcoast of each module to buffer sedimentation within the module from longshore transport (Pondella et al. 2018). Submodules were

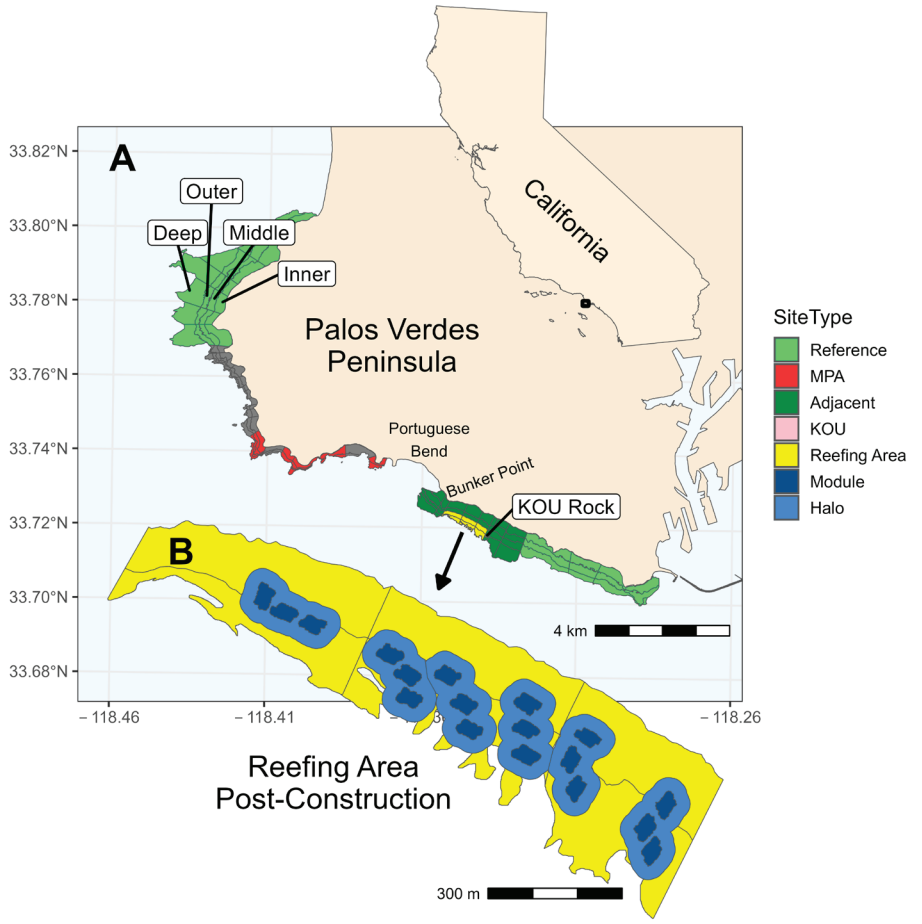


Figure 1. (A) Map of natural benthic rocky reef habitat (hard substrate covered with 1 m or less of sediment) present at Palos Verdes Peninsula, CA from the shoreline to about 30 m depth. Reef habitat is separated by depth zone (inner [0–7.5 m depth], middle [7.5–12.5 m depth], outer [12.5–17.5 m depth], deep [17.5 – about 30 m depth]) and by sampling site. Sampling sites are further categorized by site type (color) for analysis. Gray reef areas were not included in the model. The locations are also labeled for KOU Rock, a high-relief natural reef feature that served as the model for the design of the PVR modules, and for the anthropogenically induced landslides at Portuguese Bend and Bunker Point. (B) Reefing area postconstruction inset displays the 18 Palos Verdes Reef (PVR) reef modules and halos (30 m buffer around modules).

offset to maximize the ecotonal perimeter. Groups of three modules were oriented to mimic the ridges and structure of the adjacent natural reef and incorporate 10–20 m sand channels between modules (Williams et al. 2022). These sand channels were designed to allow sediment to flow through the study area. The quarry rock boulders that compose the reef are a variety of sizes ranging from 0.04 to 1.20 mt with a targeted size of 0.9 mt (about 1 m in diameter), creating varying sizes of void space that provide shelter to multiple ontogenetic life stages of fishes (Pondella et al. 2018). The reef orientations enable the modules to maximize current flow and associated zooplanktonic flux from a variety of directions (Fig. 1).



Figure 2. (A) Buried reef, example of preconstruction habitat and current halo habitat; (B) bryozoans covering Module 2A in February 2021 (8 mo postconstruction), and (C) giant kelp growing on Module 5A in November 2021 (18 mo postconstruction). Still images taken with permission from the short film "Rebirth of a Reef" by Shaun Wolfe of Shaun Wolfe Photography (Williams et al. 2022).

Fish Data Collection.—Subtidal scuba-based surveys of fishes were conducted annually at 37 sites on the Palos Verdes Peninsula from 2017 to 2023 using a standardized protocol (Pondella et al. 2019; Supplementary Table S1). Data were collected visually by teams of divers who counted, identified, and estimated individual size of fishes (TL; nearest cm) along 30 m × 2 m band transects along the reef surface, in the midwater (i.e., water column), and through the kelp canopy, when present. To avoid observer bias, the senior author conducted about 95% of the benthic fish transects including all of the benthic PVR transects. Within each site, four haphazardly located transects were conducted at each of four depth strata

(or "zones"): inner (0–7.5 m depth), middle (7.5–12.5 m depth), outer (12.5–17.5 m depth), deep (17.5 – about 30 m depth), depending on the deepest extent of rocky reef habitat present at the site. Annual survey data were categorized into two eras: preconstruction (2017–2019) and postconstruction (2020–2023). The PVR modules were constructed between 8 May and 23 September, 2020. The footprints of the modules were sampled preconstruction in spring 2020 and then we excluded the fall 2020 PVR module surveys performed in the months just after construction, resulting in the postconstruction era for the PVR modules being from 2021 to 2023. PVR modules (or their footprints preconstruction, comparable to outer or deep depth zones) were sampled annually with the same methods and effort (i.e., four transects) as a natural reef zone. Reef halos were also surveyed, consisting of four transects per reef block (i.e., a series or grouping of three modules; Fig. 1). One transect was located immediately offshore of the deepest module; one transect was located between the inshore module and natural reef; and the final two transects were in the sand channels between the three modules. Prior to analysis, coastal pelagic taxa (e.g., *Sardinops sagax*, *Trachurus symmetricus*) were excluded from the data set as these schooling fishes have high density and biomass, which are highly variable and could bias the results, only resident reef taxa were included (Allen and Pondella 2006).

Site Types and Mapping.—Sites were categorized into seven groups for analysis (Fig. 1): Reference (natural reef sites on the Palos Verdes Peninsula that are furthest away from the reefing area and less sediment impacted), Marine Protected Area (MPA; natural reef sites inside no-take marine reserves, also less sediment impacted), Adjacent (sediment-impacted natural reef sites that are directly adjacent to the reefing area), Reefing Area (low relief sediment impacted depth zones at natural reefs that PVR modules were placed upon), Module (18 discrete quarry rock modules), and Halo (a 30 m buffer around the modules, also characterized by low-relief sediment impacted natural reefs and sand channels). The 30 m buffer for the reef halos was determined from previous regional studies (Anderson et al. 1989, Johnson et al. 1994, Pondella et al. 2006). KOU Rock, is a single high-relief natural reef feature (4.5 m vertical relief, round reef feature approx. 45 m in diameter) in an outer depth zone (Fig. 3). This feature consistently supported the highest fish biomass density on the peninsula and as the model for the design of the PVR modules (Pondella et al. 2018, Williams et al. 2022), it was analyzed separately from other natural reef sites.

Area estimates for rocky reef substrate within depth zones at natural reef sites were obtained from creation of a polygon encompassing all reef area from mean low low water (MLLW) to the extent of rocky reef present (about 30 m). All natural reefs started at MLLW. Depth contours and hard substrate (i.e., reef) extent were derived from publicly-available bathymetric data (CSUMB Seafloor Mapping Lab, SFML: https://csumb.edu/undersea/seafloor-maps/) and high-resolution bathymetric data and backscatter imagery collected using an Edgetech 6205 Multiphase Echosounder. Sites were then divided based on where biological surveys were conducted and reef habitat was separated by depth zone (inner, middle, outer, deep) from depth contours derived from the bathymetric datasets. For the extent of PVR modules themselves, the high-resolution bathymetric data and backscatter imagery for PVR were classified using an object-based approach (Kettig and Landgrebe 1976, Lacharite et al. 2015), and then manually adjusted in ArcGIS Pro to create polygons of the extent of the

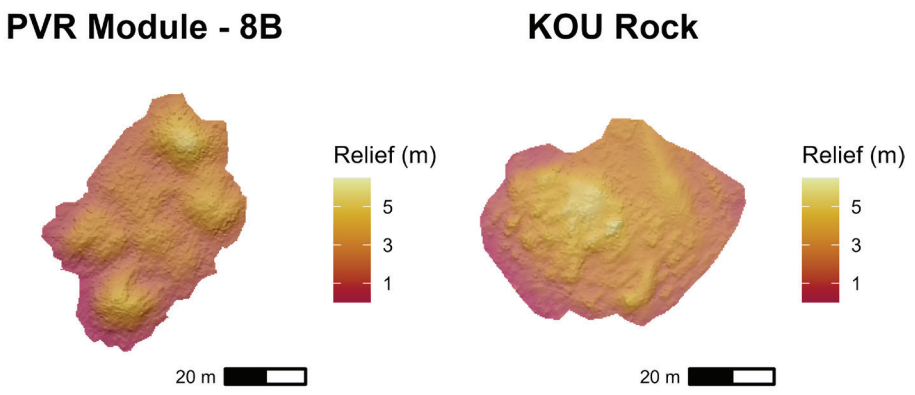


Figure 3. Sonar derived bathymetry was used to generate images of vertical relief (color) for module 8B (footprint area of 1929 m^2) and for KOU Rock (footprint area of 2139 m^2). KOU Rock is a high-relief natural reef feature adjacent to the reefing area that was used as the model for the module design (Fig. 1).

spatially distinct quarry rock modules. The 30 m halos around the modules were created using the buffer tool in ArcGIS Pro (Fig. 1).

Data Analysis.—All data analysis was completed using R (R Core Team 2024) in RStudio (RStudio Team 2024), generally using functions from tidyverse packages (Wickham et al. 2019) unless otherwise noted.

Secondary Fish Production Density Estimates.—We estimated biomass density in g m^{-2} and annual somatic production density in g m^{-2} yr^{-1} for each fish observed using standard length-weight relationships for biomass and a model developed previously for annual somatic production based on fisheries-independent density and length data of fishes from annual surveys (for more detail, *see* Miller et al. 2008, Williams et al. 2013, Claisse et al. 2014, Pondella et al. 2015). We calculated the annual somatic production rate, which is the difference between the observed biomass of an individual fish during surveys and the predicted biomass of that fish one year later times its annual survival rate. The size- and species-specific one-year increase in fish length was predicted using the Fabens version of the von Bertalanffy growth function (Haddon 2011). Biomass was calculated using species-specific weight-length relationships. The production from the proportion of fishes that were predicted to not survive the one-year time interval were not included in the production estimate. Annual survivorship rates were incorporated in the model using the size- and species-specific mortality function described in Gislason et al. (2010). Biomass and production densities were summed across species to produce a fish biomass and a fish somatic production density estimate for the assemblage. These values were then averaged across the four transects done per location annually to produce a single fish biomass or production density value for each site type, depth zone, and year.

Fish Biomass and Somatic Production Across Site Types.—Mean biomass density and secondary production density were compared pre- and postconstruction across site types and depth zones using generalized linear mixed effects models

(GLMM) with era (pre- and postconstruction), site type, and depth zone and their interactions as fixed effects, and the depth zones within each site as a random factor (since those were resampled each year) using the function glmmTMB in the R package glmmTMB (Brooks et al. 2017). We assumed a Tweedie probability distribution with a log link function as mean biomass and production replicates were non-negative, continuous, and positively skewed (Dunn and Smyth 2008, Dunn 2022). To assess model suitability and the Tweedie distribution assumption, we visually assessed distributions of production estimates for each site type, era and depth zone combination, and also checked for uniformity of the residual quantiles (observed vs expected), and for overdispersion and outliers with functions from the DHARMA package in R (Hartig 2018). Functions from the emmeans package in R (Lenth 2024) were used with the model to produce marginal means and 95% confidence intervals for each combination of factor levels with values being back transformed by the function to the original response scale. Differences were considered significant if the 95% CIs of the marginal means did not overlap. Species specific estimates are in Supplementary Table S2.

Monte Carlo Simulations of Reefing Area Absolute Fish Biomass and Annual Production.—Mapping products (areas of each polygon; Fig. 1) and estimates of fish biomass density or somatic production density means and their modelled distributions from the GLMMs were synthesized using Monte Carlo simulations (Haddon 2011) producing estimates of absolute fish biomass (kg) or annual production (kg yr^{-1}) for the reefing area pre- and postconstruction. The Monte Carlo simulations incorporate the variability associated with the fish survey data and estimated mean biomass or annual production density parameters for each era, site type, and depth zone to produce absolute estimates of annual production with 95% confidence intervals for the whole reefing area in each era (Fig. 1). In each trial of the simulation, biomass density (g 240 m^{-2}) or production density values (g 240 m^{-2} yr^{-1}) are randomly drawn (using the function rtweedie from the tweedie package in R; Dunn and Smyth 2008, Dunn 2022) from a Tweedie distribution until the sum of the annually sampled areas (four 60 m^2 transects = 240 m^2 of seafloor sampled) equaled the total polygon area for each site type and depth zone in the reefing area for each era. Variability in estimating the era, site type, and depth zone specific mean biomass or production densities was accounted for by randomly drawing a mean biomass or production value (using the R function rtruncnorm) from their own truncated (i.e., no negative values) normal distributions according to the parameter estimates and their associated standard errors obtained from fitting the Tweedie distribution to the available empirical biomass or production data. A mean and 95% confidence interval of absolute biomass or annual somatic production for each era were then calculated from 5000 simulation trials.

Results

Annual Fish Biomass and Production Density Comparison.—Preconstruction, across site types and depth zones (Fig. 1), the highest mean fish biomass and annual somatic production densities were primarily observed in the outer and deep depth zones, with lower mean values in the inner and middle depth zones (Figs. 4–6). In these depth zones, a similar suite of species were the major contributors

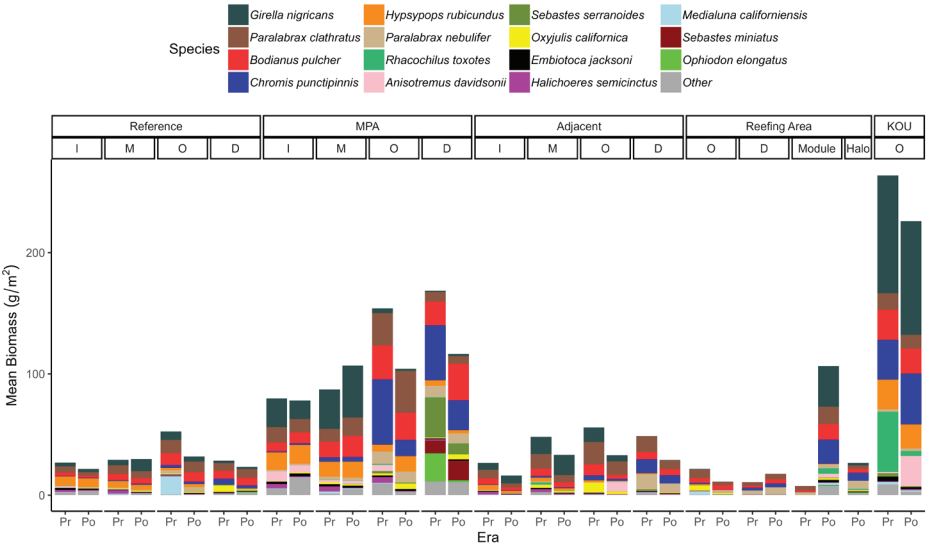


Figure 4. Species-specific contributions to mean biomass (g m^{-2}) estimates by site type (top), depth zone (top; I: Inner, M: Middle, O: Outer, D: Deep, and Module or Halo) and era (*x*-axis; Pr: Preconstruction, Po: Postconstruction). Species in legend are ordered based on their overall contribution to biomass averaged across all categories, and those that contribute <1% to this overall average are combined into Other.

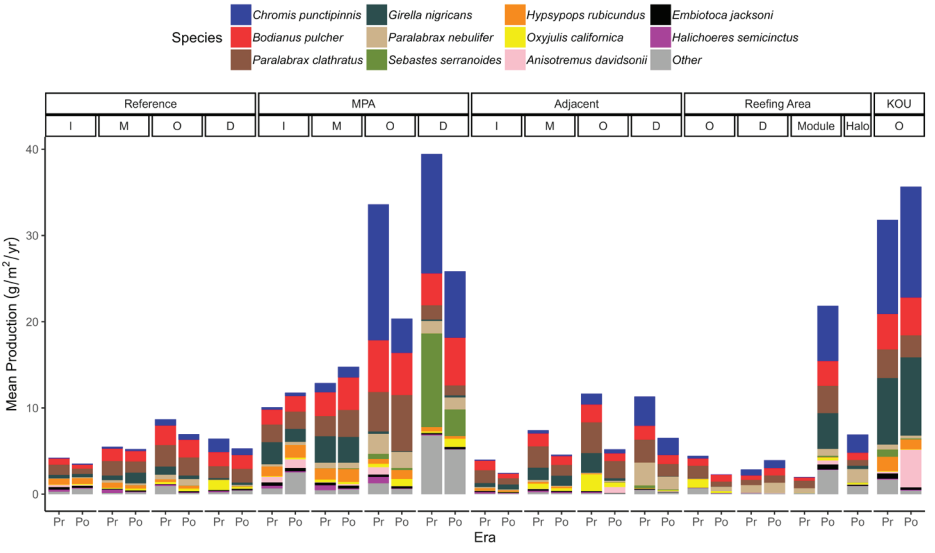


Figure 5. Species-specific contributions to mean production (g m^{-2} yr^{-1}) estimates by site type (top), depth zone (top; I: Inner, M: Middle, O: Outer, D: Deep, and Module or Halo) and era (*x*-axis; Pr: Preconstruction, Po: Postconstruction). Species in legend are ordered based on their overall contribution to production averaged across all categories, and those that contribute <1% to this overall average are combined into Other.

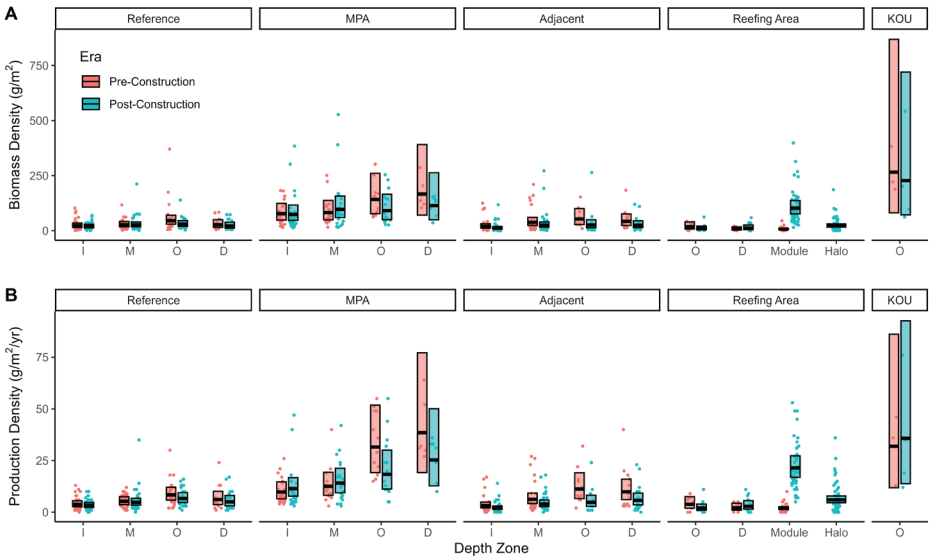


Figure 6. (A) Mean fish biomass density (g m^{-2}) and (B) somatic production density (g m^{-2} yr^{-1}) by site type (top labels) and depth zone (*x*-axis; I: Inner, M: Middle, O: Outer, D: Deep, and Module or Halo) and Era (color). Points are means across transects of biomass density or production density summed across species for a given site, site type, depth zone, and year. Means (line) and 95% CI (box, by era) are reported from GLMM assuming a Tweedie probability distribution with a random factor for the resampled (yearly) Site, Site Type, and Depth Zone.

to biomass and production across all site types with few exceptions (Figs. 4 and 5; Supplementary Table S2). The highest mean fish biomass and production densities were generally in the MPAs (biomass: outer 154 g m^{-2}, deep 169 g m^{-2}; production: outer 34 g m^{-2} yr^{-1}, deep 39 g m^{-2} yr^{-1}), followed by the Adjacent (biomass: outer 56 g m^{-2}, deep 49 g m^{-2}; production: outer 12 g m^{-2} yr^{-1}, deep: 11 g m^{-2} yr^{-1}), and then the Reference sites (biomass: outer 53 g m^{-2}, deep 29 g m^{-2}; production: outer 9 g m^{-2} yr^{-1}, deep 6 g m^{-2} yr^{-1}) site types (Fig. 6, Supplementary Table S2). An exception was KOU Rock, adjacent to the reefing area (Figs. 1 and 3), which had a similar preconstruction mean fish biomass density (264 g m^{-2}) and rate of production density (32 g m^{-2} yr^{-1}) as the MPA sites. Preconstruction, the outer and deep depth zones surveyed at the sediment-impacted low-relief natural reefs in the reefing area had the lowest levels of mean fish biomass and production densities (biomass: outer 22 g m^{-2}, deep 11 g m^{-2}; production: outer 5 g m^{-2} yr^{-1}, deep 3 g m^{-2} yr^{-1}), with the lowest values being within the footprints of where the PVR modules were built (biomass 8 g m^{-2}; production 2 g m^{-2} yr^{-1}; Figs. 4 and 6, Supplementary Table S2).

Between eras (from pre- to postconstruction), mean fish biomass and production densities generally decreased regionally, including in the outer and deep depth zones at the Adjacent, MPA, and Reference site types, while increasing significantly on the PVR module and Halo habitats (Table 1, Fig. 6). We observed changes in production and biomass across all depth zones and treatments (Fig. 6, Supplementary Table S2). Mean fish biomass and production densities of sites directly adjacent to the reefing area (Adjacent) declined postconstruction across all depth zones, ranging between −31% to −41% in biomass and −39% to −55% in production (Fig. 6, Supplementary Table S2), while also declining in the outer and deep depth zones in the MPAs

Table 1. Palos Verdes Reef (PVR) reefing area habitat extent (m^2), fish biomass density (g m^{-2}) and total fish biomass (kg), fish somatic production density (g m^{-2} yr^{-1}), and total annual production (kg yr^{-1}) estimates preconstruction (pre) and postconstruction (post).

Habitat	Area m^2	Biomass					Production				
		Density g m^{-2} (95% CI)		Total kg (95% CI)			Density g m^{-2} yr^{-1} (95% CI)		Total kg yr^{-1} (95% CI)		
		Pre	Post	Pre	Post	% Change	Pre	Post	Pre	Post	% Change
Module	32,158	6.9 (4.6–10.4)	102.2 (76.3–136.9)	223 (128–326)	3,286 (2,276–4,297)	1,372%	1.8 (1.3–2.6)	21.5 (16.9–27.3)	59 (38–82)	690 (516–871)	1,062%
Halo	102,517	6.9 (4.6–10.4)	22.5 (16.3–31.1)	711 (423–1,009)	2,303 (1,540–3,086)	224%	1.8 (1.3–2.6)	6.0 (4.6–7.9)	188 (123–255)	618 (451–791)	229%
Outer	138,029	17.9 (8.0–39.9)	9.5 (4.3–20.7)	2,486 (605–4,435)	1,314 (321–2,377)	−47%	3.8 (1.9–7.5)	2.0 (1.0–3.9)	525 (166–876)	278 (96–467)	−47%
Deep	79,645	7.3 (3.1–17.3)	11.7 (5.3–25.8)	584 (113–1,082)	947 (221–1,671)	62%	2.1 (1.0–4.4)	2.9 (1.5–5.6)	170 (53–293)	230 (79–388)	35%
Total	352,350			4,005 (2,065–6,043)	7,850 (6,101–9,700)	96%			943 (551–1,328)	1,816 (1,472–2,169)	93%

Notes: Biomass density and production density estimates and 95% CIs are from their respective GLMM models (Fig. 5) and the total biomass and production estimates and their 95% CIs are from the Monte Carlo simulation (Fig. 7). The reefing area includes the natural (often buried by <1 m sediment) reefs in the outer and deep depth zones, the PVR modules, and the PVR (30 m) halo habitats. Totals for the reefing area (summed across habitats) and the Post-Pre percent change total biomass and total production are also reported.

(biomass: outer –32%, deep –31%; production: outer –39%, deep –35%) and the Reference sites (biomass: outer –39%, deep –18%; production: outer –24%, deep: –17%, Supplementary Table S2). KOU Rock (Fig. 1) was again an exception as fish production density there increased slightly (12%) postconstruction (post 35.7 g m^{-2} yr^{-1}) to a level higher than the means within the outer and deep MPA sites (Fig. 6, Supplementary Table S2). Postconstruction on the PVR modules, there were significant (no overlap of 95% CIs) increases in fish biomass density (1293%, pre 8 g m^{-2}, post 107 g m^{-2}) and production density (1001%, pre 2 g m^{-2} yr^{-1}, post 22 g m^{-2} yr^{-1}) to a level that was similar to the postconstruction fish biomass and production densities in similar depth zones within the MPAs (biomass: outer 104 g m^{-2}, deep 117 g m^{-2}; production: outer 20 g m^{-2} yr^{-1}, deep 26 g m^{-2} yr^{-1}; Fig. 6, Supplementary Table S2). For individual PVR modules postconstruction mean values ranged from 52.4 to 199.4 g m^{-2} for biomass density (659% to 2790% increase above preconstruction mean) and 14.3 to 35.7 g m^{-2} yr^{-1} for production density (694% to 1883% increase above preconstruction mean; Fig. 6). The PVR Halo habitat (i.e., the 30 m buffer surrounding the modules; Fig. 1) exhibited a significant 326% increase in biomass density (pre 6.9 g m^{-2}, post 22.5 g m^{-2}) and a 323% increase in production (pre 1.8 g m^{-2} yr^{-1}, post 6.0 g m^{-2} yr^{-1}; Table 1).

On the PVR modules, fish biomass and production increased driven by large (hundreds to thousands of percent) increases in abundance of multiple species, and by additional species that had not previously been observed in their footprint. Overall, on the PVR Modules 76% of both the fish biomass and somatic production densities postconstruction were contributed by four species: *Chromis punctipinnis* (biomass: 20 g m^{-2}, 19%; production: 6 g m^{-2} yr^{-1}, 29%), *Girella nigricans* (biomass: 34 g m^{-2}, 32%; production: 4 g m^{-2} yr^{-1}, 19%), *Paralabrax clathratus* (biomass: 14 g m^{-2}, 13%; production: 3 g m^{-2} yr^{-1}, 15%), and *Bodianus pulcher* (biomass: 13 g m^{-2}, 12%; production: 3 g m^{-2} yr^{-1}, 13%; Figs. 4 and 5, Supplementary Table S2). *Girella nigricans* had not been observed preconstruction on the module footprint surveys, but was present in the restoration area, along with three other species that made notable (>2%) contributions to the postconstruction biomass and production densities: *Rhacochilus toxotes* (biomass: 5 g m^{-2}, 4%; production: 1 g m^{-2} yr^{-1}, 3%), *Embiotoca jacksoni* (biomass: 2 g m^{-2}, 2%; production: 2 g m^{-2} yr^{-1}, 2%), *Anisotremus davidsonii* (biomass: 3 g m^{-2}, 3%; production: 0.5 g m^{-2} yr^{-1}, 2%).

Change in Absolute Fish Biomass (kg) and Annual Production (kg yr^{-1}) Across the Reefing Area.—For the 35.2 ha reefing area, the absolute amount of fish biomass (kg) and total somatic production (kg yr^{-1}) both approximately doubled postconstruction (Table 1, Fig. 7B and D). Absolute biomass significantly increased by 96% (pre: 4005 kg, 95% CI 2065–6043, post: 7850 kg, 95% CI 6101–9700, simulation-based P-value < 0.003) and absolute somatic production significantly increased by 93% (pre: 943 kg yr^{-1}, 95% CI 551–1328, post: 1816 kg yr^{-1}, 95% CI 1472–2169, simulation-based P-value < 0.001). Within the reefing area, preconstruction 20 fish species contributed to these estimates, compared to 38 species postconstruction (Supplementary Table S2). The absolute fish biomass and somatic production rate for all the PVR modules increased 1372% and 1062%, respectively, compared to their footprints (Supplementary Table S2). While fish biomass and production were concentrated on the PVR modules (Fig. 7A and C), the increase in biomass and production contributed by the 30 m halos surrounding the modules resulted in an

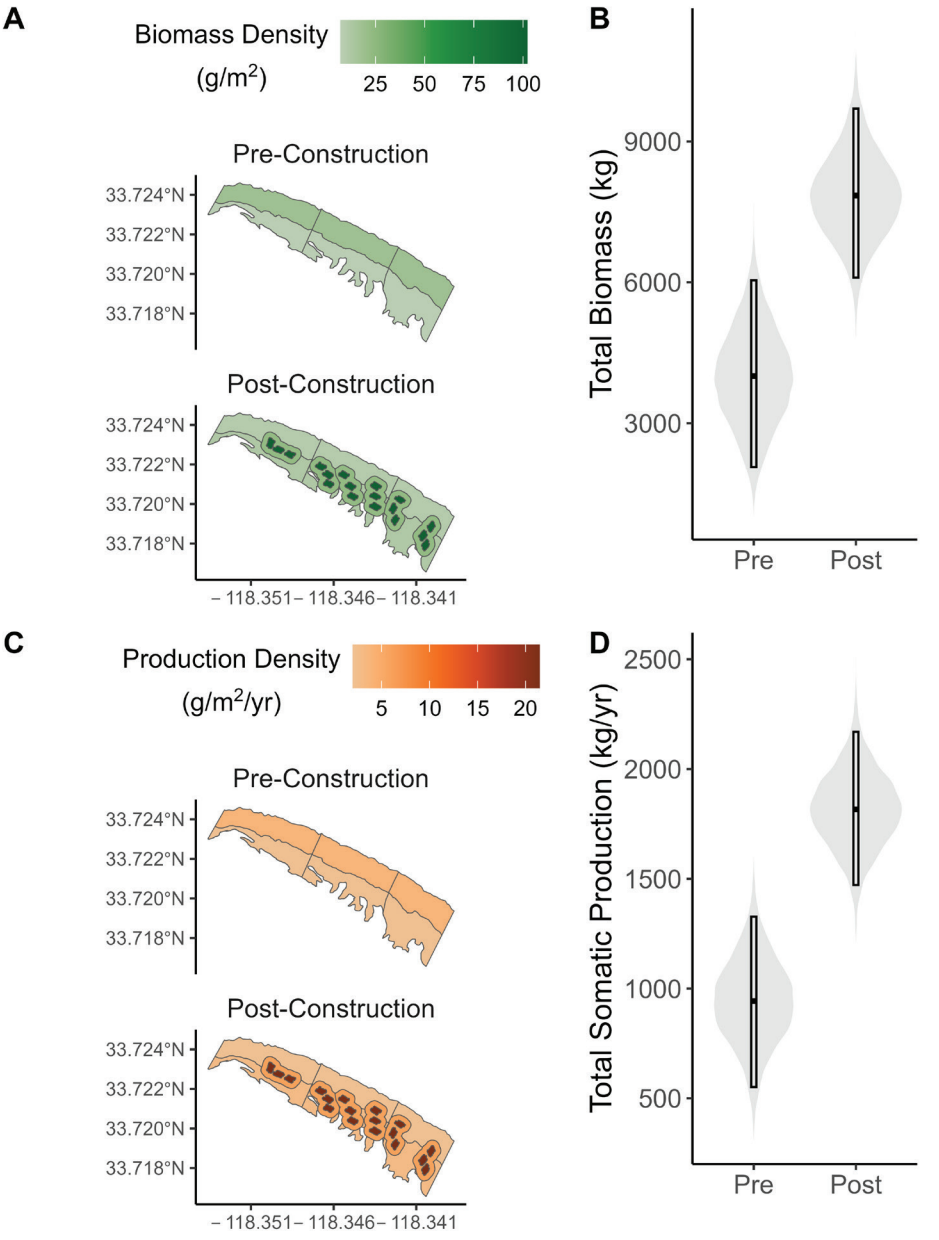


Figure 7. Maps of (A) mean biomass density (g m^{-2}) and (C) mean somatic production density (g m^{-2} yr^{-1}) estimates from GLMMs (Fig. 5) across the reefing area which includes the natural (often buried by <1 m sediment) reefs in the outer and deep depth zones, the PVR modules and the PVR (30m) halo habitat. Monte Carlo simulation outputs of summed (B) total biomass (kg) and (D) total somatic production (kg yr^{-1}) across the reefing area. The distributions of simulation replicates (n = 5000) for Preconstruction and Postconstruction are displayed as violin plots (light gray), along with the mean and 95% confidence limits of the replicates. There were significant increases in total biomass (P = 0.003) and total somatic production (P < 0.001) postconstruction (simulation-based P-values, proportion of 5000 simulations where Post < Pre).

almost equivalent contributions of annual production as the modules themselves (modules: 690 kg yr^{-1}, 95% CI 516–871, halos: 618 kg yr^{-1}, 95% CI 451–791; Table 1). Meanwhile, the remaining habitat in the reefing area exhibited relatively small changes in the absolute fish biomass and annual production rate postconstruction, depending on the depth zone (Table 1, Fig. 7A and C).

Discussion

In a first for California, a subtidal offshore rocky reef was restored in situ. 3.2 ha of strategically-placed granitic quarry rock minimized the effects of persistent sedimentation and associated turbidity, rapidly becoming a vibrant rocky-reef. Preconstruction, the sediment-impacted natural reefs in the restoration area had the lowest levels of mean fish production density in the region. PVR was built in the outer and deep depth zones, which generally had the highest mean annual fish production densities across all site types throughout the Palos Verdes Peninsula. Further, the reef design included maximizing ecotonal and halo habitats, utilizing the connectivity among the new reef modules and existing natural reefs to maximize fish production over a 35.2 ha restoration area. This enabled immediate connectivity throughout the existing reefs and the restoration habitat. As anticipated, there was an initial colonization by demersal fishes, resulting in a 47% decrease in somatic biomass and production in the immediate surrounding habitat, indicating preference for the new habitat across all taxa indicative of attraction to the new habitat. Biomass and production in the footprints of the new modules increased 1372% and 1072%, respectively, during the three years postconstruction in part due to this redistribution of fishes. Similar increases in biomass density 6.9 to 102.2 g m^{-2} and production density 1.8 to 21.5 g m^{-2} yr^{-1} occurred on the PVR modules. The initial redistribution was more than offset by the gains across the restoration area. In part, this was due to a substantial production increase in the newly created Halos (224% increase in biomass, 711 to 2303 kg and 229% increase in production, 188 to 618 kg yr^{-1}). Accounting for the changes in fish production in the halo surrounding an artificial reef module was impactful. Halos, which cover approximately three times the seafloor area as the modules, contributed almost as much absolute annual fish production (618 kg yr^{-1}) as the PVR modules themselves (708 kg yr^{-1}). The fishes detected occupying the Halos were primarily the reef taxa (e.g., *C. punctipinnis*, *Paralabrax nebulifer*, *B. pulcher*, and *P. clathratus*) associated with the new modules and surrounding natural reefs (Supplementary Table S2). In the surrounding offshore habitats (deep depth zone) there was also a 62% increase in biomass (584 to 947 kg) and a 35% increase in production (170 kg yr^{-1} to 230 kg yr^{-1}). Overall, there was a 96% increase in biomass (4005 to 7850 kg) and a 93% increase in somatic production (943 to 1816 kg yr^{-1}) for the entire 35.2 ha restoration area, demonstrating that carrying and productive capacity almost doubled during the initial 3-yr period postconstruction (Table 1, Fig. 7). The modules had a dramatic localized (PVR + Halos) effect, and this effect spilled over into the entire restoration habitat.

Fish biomass and secondary production estimates for marine systems, and more specifically for artificial reef habitats, vary considerably. In California, secondary production on deep (>30 m) rocky reefs was 4.4–22.4 g m^{-2} yr^{-1} (Claisse et al. 2014), on a mature artificial reef was 66.5 g m^{-2} yr^{-1} (Johnson et al. 1994), and on oil platforms was 104.7–886.8 g m^{-2} yr^{-1} (Claisse et al. 2014). In an assessment of three taxa, oil

platforms in southeastern Australia had an average production of 82 g m^{-2} yr^{-1} (Birt et al. 2024). Previously reviewed global fish production rates for nearshore marine ecosystems varied between 0.9 g m^{-2} yr^{-1} and 74.2 g m^{-2} yr^{-1}, with most estimates from bay and estuarine habitats (Claisse et al. 2014) and the highest from a coral reef in Moorea (Galzin 1987). More recently, fish production was estimated for three coral reef systems at 69 to 486 g m^{-2} yr^{-1} (Morais et al. 2021). These are some of the highest estimates globally, and similar to those from the southern California oil platforms. That method, which anchors growth model parameter estimates to a taxon's maximum recorded length, likely results in relatively higher production rates, although the magnitude of the difference is unknown (Morais and Bellwood 2018). Our model uses empirically derived von Bertalanffy growth function parameters from the literature, which generally result in fish growth reaching asymptote at relatively shorter lengths. In Sydney, Australia, an artificial reef had a production rate of 384 g m^{-2} yr^{-1} (Smith et al. 2016). However, they included visitor (i.e., pelagic) species not included in this analysis or in our platform study, making direct comparisons challenging since we specifically excluded these taxa (Claisse et al. 2014, Pondella et al. 2015). Biomass density for nearshore rocky reef fishes across the Southern California Bight varies considerably (8.0–219.5 g m^{-2}) with an average of 71.1 g m^{-2} (Pondella et al. 2019). The new PVR modules have an average fish biomass density of 102.2 g m^{-2} and secondary production density of 21.5 g m^{-2} yr^{-1}, which indicated the estimates are reasonable and that there is potentially more capacity in this system as the reefs mature.

The overall magnitude of this restoration effort is greater than the absolute values reported; during this three-year period postconstruction, mean fish production and biomass densities generally decreased across the entire region (Fig. 6, Supplementary Table S2). The PVR modules had similar levels of mean fish biomass density (g m^{-2}) and annual somatic production density (g m^{-2} yr^{-1}) to the nearby marine reserve (in place since 2012) sites at similar depth zones. KOU Rock, the high relief natural reef in the outer depth zone (which decreased overall) did not exhibit this loss, indicating that the high relief habitat is in fact limiting and had a similar mean rate of fish production density as the MPA sites preconstruction. Both KOU Rock and PVR are open to fishing, therefore the relative production rate would be higher without the offset of fishing mortality. Against the backdrop of declining productivity, PVR and the reef it was modeled after, KOU, are distinct.

Major contributors to annual fish production included a planktivore *C. punctipinnis*, a herbivore/omnivore *G. nigricans*, and mesocarnivores (*P. clathratus* and *P. nebulifer*) and a primarily invertivore (*B. pulcher*). The three carnivores are all important fishery species. The design criteria of the modules, high relief with 1 mt granitic quarry rock, specifically enhance sheltering and nesting sites for *C. punctipinnis* as well as the maximization of current flow through the system. Similarly, its location exhibits relatively strong tidal flushing and close proximity to the 200 m isobath an indication of cool nutrient rich water (Pondella et al. 2019, Zellmer et al. 2019, Williams et al. 2022). Planktivorous fishes were observed at the highest densities on higher relief submodules located at the offshore ends of modules (Sturges and Claisse 2024). Optimal habitat plus physical oceanographic forcing and resulting planktonic flux appears to have resulted in the immigration of these planktivores to the high relief and complex modules at PVR as designed. These planktivores provide a critical trophic link and fuel reef primary production with

their excretion of inorganic nitrogen, essential for the growth of macroalgae (Bray et al. 1981, 1988, Shrestha et al. 2023, 2024). The role of planktivores as a central driver in reef fish production is well-documented (Morais and Bellwood 2019, Morais et al. 2021, Pinnegar and Polunin 2006). This reef design, with high vertical relief, clearly increases the dome effect of planktivores foraging in the water column above the reef (Pondella et al. 2022), and the planktivores responded with the highest production, a critical design element to maximize primary and secondary production through benthic-pelagic coupling. Reefs (MPAs, KOU and PVR) that had high planktivore production had the highest overall production in the region.

Some previous reefing projects have produced localized to regional increases in fish production (Polovina and Sakai 1989, Steimle et al. 2002, Santos et al. 2011, Roa-Ureta et al. 2019). This present study expands upon these examples. First, we optimized spacing criteria: modules placed 10 m apart and about 10 m from the natural reefs allow connectivity among these reefs (Pondella et al. 2006), extending the natural reef offshore. The redistribution of fishes in the outer depth strata demonstrated a preference for and assessment of preferred habitat (in the manner of Warner 1990) compared to the reefs they had previously occupied. The creation and optimization of reef halos and associated ecotones enhance production as fishes forage in these areas facilitating positive feedback among the softbottom and hardbottom communities. Further, accounting for the production of fishes from the halo habitat, which had almost as much absolute fish production annually as the modules themselves, provided a more complete assessment of the overall impact of the reef restoration effort. Module construction specifically mimicked a highly productive natural reef proximate to the restoration site (KOU). Considering the engineering design criteria PVR performed similar to KOU (Pondella et al. 2018; Fig. 3). Through small-scale species-specific habitat use patterns (Sturges and Claisse 2024), the heterogeneous design of the PVR with varying levels of vertical relief across submodules and their adjacent ecotones (Fig. 2) facilitated increased biodiversity and ontogenetic habitat use patterns. Further, this design created habitat that was resistant to the benthic deleterious effects of persistent sedimentation, we observed no burial or scour, as the surrounding ecosystem has reduced production due to these documented impacts. By optimizing these reefing metrics, we were able to measure a nearly doubled carrying and productive capacity over the entire 35.2 ha restoration area 3 yr postconstruction, with the justified expectation that this measured capacity will increase over time, an indication of restoration success.

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